Throughput, 200 PU200 ttbar events (RelVal files 3+4), 4 threads, arm (cmssw-el9-native-1)

quantity	0	1	2
mode	payload	emu v1 (HEAD fc495c2, recorded 2026-09-27)	emu v2 (fix round)
events	200	200	200
steady s/event (begin-processing spacing, events > 20)	2.36	8.23	2.85
steady events/s	0.424	0.1215	0.3504
first 20 events [s]	46.5	160.5	55.3
job wall [s]	519.2	1705.5	613.2
job CPU user+sys [s]	1570.7	6407.9	2054.0
CPU/event (job avg) [s]	7.85	32.04	10.27
peak RSS [GB]	6.94	7.0	7.12
FastTimer event loop real [s]	1941.3	6632.6	2324.1
ratio to payload	1.0	3.487	1.208

Top-12 modules per mode (FastTimerService, all events)

mode	module	type	real/event [ms]	cpu/event [ms]	events
emu v1 (HEAD fc495c2, recorded 2026-09-27)	spixSmartPixelsRecHits	SmartPixelsRecHitProducer	24278.8	24211.2	200
payload	other	other	3711.8	2528.2	200
emu v1 (HEAD fc495c2, recorded 2026-09-27)	source	PoolSource	3125.3	2558.6	200
emu v1 (HEAD fc495c2, recorded 2026-09-27)	NANOAODSIMoutput	NanoAODOutputModule	2804.7	2760.7	200
emu v2 (fix round)	NANOAODSIMoutput	NanoAODOutputModule	2757.3	2676.0	200
emu v2 (fix round)	other	other	2678.9	1976.4	200
payload	NANOAODSIMoutput	NanoAODOutputModule	2283.6	2214.4	200
emu v2 (fix round)	source	PoolSource	2278.3	1756.8	200
emu v2 (fix round)	spixSmartPixelsRecHits	SmartPixelsRecHitProducer	2272.3	2263.9	200
payload	source	PoolSource	1912.6	1377.7	200
emu v1 (HEAD fc495c2, recorded 2026-09-27)	other	other	993.3	597.7	200
payload	l1tTTTracksFromExtendedTrackletEmulation	L1FPGATrackProducer	559.0	555.3	200
emu v2 (fix round)	l1tTTTracksFromExtendedTrackletEmulation	L1FPGATrackProducer	552.3	549.6	200
emu v1 (HEAD fc495c2, recorded 2026-09-27)	idle	idle	549.8	352.4	200
emu v1 (HEAD fc495c2, recorded 2026-09-27)	l1tTTTracksFromExtendedTrackletEmulation	L1FPGATrackProducer	543.4	541.6	200
payload	idle	idle	303.7	-393.5	200
emu v2 (fix round)	idle	idle	284.9	434.2	200
emu v1 (HEAD fc495c2, recorded 2026-09-27)	cleanup	cleanup	260.4	259.3	200
emu v2 (fix round)	cleanup	cleanup	221.1	219.6	200
payload	spixSmartPixelsRecHits	SmartPixelsRecHitProducer	198.9	197.4	200
payload	cleanup	cleanup	172.5	171.2	200
emu v2 (fix round)	spixPixelClusters	SiPixelClusterProducer	101.2	98.5	200
payload	spixPixelClusters	SiPixelClusterProducer	100.4	97.0	200
emu v1 (HEAD fc495c2, recorded 2026-09-27)	spixPixelClusters	SiPixelClusterProducer	99.5	96.8	200
emu v1 (HEAD fc495c2, recorded 2026-09-27)	l1tTTTracksFromTrackletEmulation	L1FPGATrackProducer	97.6	97.2	200
payload	l1tTTTracksFromTrackletEmulation	L1FPGATrackProducer	96.3	95.7	200
emu v2 (fix round)	l1tTTTracksFromTrackletEmulation	L1FPGATrackProducer	94.3	93.6	200
emu v1 (HEAD fc495c2, recorded 2026-09-27)	l1tSmartPixelsTrackProducerExtendedWdigiRefitAAAA	L1SmartPixelsTrackProducer	75.9	75.7	200
emu v2 (fix round)	l1tSmartPixelsTrackProducerExtendedWdigiRefitAAAA	L1SmartPixelsTrackProducer	67.7	67.3	200
emu v1 (HEAD fc495c2, recorded 2026-09-27)	spixPixelRecHits	SiPixelRecHitConverter	66.0	64.9	200
emu v2 (fix round)	spixPixelRecHits	SiPixelRecHitConverter	63.6	62.6	200
payload	spixPixelRecHits	SiPixelRecHitConverter	62.6	61.5	200
payload	l1tSmartPixelsTrackProducerExtendedWdigiRefitAAAA	L1SmartPixelsTrackProducer	61.5	61.2	200
payload	l1tSmartPixelsTrackProducerWdigiRefitAAAA	L1SmartPixelsTrackProducer	58.0	57.5	200
emu v1 (HEAD fc495c2, recorded 2026-09-27)	l1tPh3SmartPixelsClusterTable	L1SmartPixelsClusterTableProducer	57.8	57.1	200
emu v2 (fix round)	l1tSmartPixelsTrackProducerWdigiRefitAAAA	L1SmartPixelsTrackProducer	55.8	55.6	200

SmartPixels modules (spix*) per mode

mode	module	type	real/event [ms]	cpu/event [ms]	events
payload	spixPixelClusters	SiPixelClusterProducer	100.4	97.0	200
payload	spixPixelRecHits	SiPixelRecHitConverter	62.6	61.5	200
payload	spixSmartPixelsRecHits	SmartPixelsRecHitProducer	198.9	197.4	200
emu v1 (HEAD fc495c2, recorded 2026-09-27)	spixPixelClusters	SiPixelClusterProducer	99.5	96.8	200
emu v1 (HEAD fc495c2, recorded 2026-09-27)	spixPixelRecHits	SiPixelRecHitConverter	66.0	64.9	200
emu v1 (HEAD fc495c2, recorded 2026-09-27)	spixSmartPixelsRecHits	SmartPixelsRecHitProducer	24278.8	24211.2	200
emu v2 (fix round)	spixPixelClusters	SiPixelClusterProducer	101.2	98.5	200
emu v2 (fix round)	spixPixelRecHits	SiPixelRecHitConverter	63.6	62.6	200
emu v2 (fix round)	spixSmartPixelsRecHits	SmartPixelsRecHitProducer	2272.3	2263.9	200

SmartPixelsRecHitProducer emu steps (wall time summed over the 4 streams)

mode	step	total [s]	per event [ms]
emu v1 (HEAD fc495c2, recorded 2026-09-27)	helix (class-T test)	42.54	212.72
emu v1 (HEAD fc495c2, recorded 2026-09-27)	CNN (ONNX)	2926.47	14632.34
emu v1 (HEAD fc495c2, recorded 2026-09-27)	donor kNN	1707.12	8535.59
emu v1 (HEAD fc495c2, recorded 2026-09-27)	dump kNN	75.5	377.52
emu v1 (HEAD fc495c2, recorded 2026-09-27)	draw+bookkeeping	36.36	181.82
emu v1 (HEAD fc495c2, recorded 2026-09-27)	one-time setup (ONNX sessions, library load, KD-tree builds)	5.04	nan
emu v2 (fix round)	helix (class-T test)	57.06	285.29
emu v2 (fix round)	CNN inputs	2.92	14.62
emu v2 (fix round)	CNN (ONNX)	235.48	1177.42
emu v2 (fix round)	A1 (inputs + ONNX)	23.44	117.18
emu v2 (fix round)	dump kNN	58.77	293.83
emu v2 (fix round)	response+bookkeeping	29.24	146.18
emu v2 (fix round)	one-time setup (ONNX sessions, library load, KD-tree builds)	0.21	nan